

Remote Sensing and Machine Learning Based Above Ground Biomass Carbon Estimation of Dry Tropical Forests in Data Limited Regions

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Abstract

Remote sensing imagery and machine learning algorithms are increasingly applied to quantify vegetation characteristics such as biomass and carbon density owing to the recent growing availability of satellite data and cloud-based computing technologies. In this study, we employ machine learning models on multispectral band reflectance data extracted from Landsat 8 satellite imagery, along with seven vegetation indices, to estimate the above-ground biomass (AGB) of a dry afromontane forest in Tigray, Northern Ethiopia. We apply several Machine learning algorithms including Light Gradient Boosting (LightGB), Extreme Gradient Boosting Regressor (XGB), Random Forest (RF), and Support Vector Regression (SVR) to accurately estimate the AGB of the Desa'a dry afromontane forest and evaluate the performance of the models. In doing so, we divided the training dataset into five sections with the K-fold cross-validation (K-fold CV) technique and selected the optimal subset of features for each estimator. The efficiency of the models was then evaluated and compared based on metrics such as mean absolute error (MAE), root mean square error (RMSE), and cross-validated root mean square error (CV-RMSE). Hyperparameter tuning in ensemble models significantly enhances predictive performances. Notably, RF, LightGB, and XGB with hyperparameter tuning lowers the RMSE values and exhibited superior predictive capabilities compared to regular gradient boosting model, which suffers from overfitting. In addition, regularization techniques are implemented on regression-based machine learning models to identify robust patterns in the data, prevent overfitting, and improve the generalization capabilities of the models in the estimation. The research findings are crucial for programs like REDD+ and other carbon markets, relying on accurate monitoring for tropical forest-based CO₂ emissions reduction or increased removals of atmospheric CO₂. This is particularly true in developing countries where financial constraints and security impediments limit the feasibility of field surveys and data collection.

Key words: Carbon Credit, Climate Change, Dry Afromontane Forests, Hyperparameters Tuning, Variable Importance

1. Introduction

Tropical forests, comprising approximately 45% of global forest cover (FAO, 2020) play a critical role in reducing the ever-increasing global temperature by absorbing carbon dioxide (CO₂) from the atmosphere and storing it as carbon in their biomass, particularly, aboveground biomass (AGB) as well as through their capacity to make clouds, humidify the atmosphere, and emit cooling compounds (Kreier, 2022). Nevertheless, tropical deforestation and forest degradation are among the primary sources of CO₂ emissions (Baccini et al., 2017). Consequently, in the effort towards mitigating global climate change, there has been considerable interest in using tropical forestry-related measures to reduce both the CO₂ concentrations and emissions to the atmosphere. Halting tropical deforestation and forest degradation, promoting forest conservation and sustainable tropical forest management are suggested as a climate change mitigation mechanism. REDD+ (Reducing emissions from deforestation and forest degradation) in developing countries, a climate change mitigation framework created by the UNFCCC's COP (United Nations Framework Convention on Climate Change's Conference of the Parties), primarily addresses this goal (UNFCCC, 2010). REDD+ is now prominently included in the Paris Agreement, which aims to keep global temperature increase below 2°C compared to pre-industrial levels (Masson-Delmotte et al., 2018). REDD+ is also directly related to Goal 13 and Goal 15 of the Sustainable Development Goals (SDGs), which highlight the need to take urgent action to combat climate change and its impacts and to halt global deforestation, respectively. The New York Declaration on Forests is another global effort that aims to end deforestation completely by 2030.

Owing to the growing global commitments related to climate change and forests, the demand for data and information about the earth's forests, particularly tropical forests, has been growing. All commitments, particularly performance-based mitigation efforts such as the REDD+ require credible and consistent monitoring systems to assess progress toward forests and climate-related targets (Petersen et al., 2018). Under the REDD+ scheme, governments in developing tropical countries would receive compensation based on achieved reductions of forest-based CO₂ emissions or increased sequestration of carbon. Hence, the implementation of REDD+ and other voluntary carbon markets that offer financial incentives for forest-based climate change mitigation depends largely on reliable monitoring, reporting, and verification (MRV) protocols of carbon sequestration and storage. Moreover, MRV is the key to accessing climate finance. However, many low-income countries like Ethiopia (the case country) lack the

capacity to develop MRV protocols themselves, and thus their access to global carbon finance remains limited.

Remote sensing (RS) technologies are important data sources to fill in the information gaps on the existing biomass/carbon monitoring, and associated limitations, particularly in developing tropical countries where monitoring is most needed (Gyamfi-Ampadu & Gebreslasie, 2021; Murrins Misiukas et al., 2021). Rapid advances in remote sensing technologies, cloud computing platforms (e.g., Google Earth Engine (GEE)), and machine learning and deep learning algorithms as well as increases in both the availability and diversity of satellite images made it possible to estimate and monitor changes in forests as well as their attributes and ecosystem services faster, cheaper, and at higher spatial, temporal, and spectral resolutions. Though there have been recent developments in airborne and ground-based sensors with advanced 3D measurements, space-borne RS remains a widely popular forest monitoring method (Petersen et al., 2018). Spaceborne sensors offer unique advantages due to their high temporal resolution, making monitoring of forests and their attributes as well as tracking their dynamics in data-deficient tropical countries possible (Mathur & Bhattacharya, 2023). Their increasing availability is another important factor that makes spaceborne RS more relevant in low-income countries like Ethiopia. Moreover, there is a growing interest in promptly and cost-effectively determining forest AGB using methods that are less destructive than the traditional field measurement techniques that involve cutting and weighing trees to calculate AGB (Silveira et al. 2019). While this approach provides precise AGB estimates, it is labor-intensive, expensive, and time-consuming, making it unsuitable for large-scale applications (Li et al., 2013). In contrast, RS techniques offer a prompt and cost-effective alternative that can furnish valuable AGB information across large areas, and promptly (Li et al., 2020). Previous studies (for example Deng et al., 2014; Li et al., 2020) highlighted the strong correlation between RS data and AGB, indicating that it can be employed for accurate biomass prediction. As a result, this study utilizes optical satellite imagery and state-of-the-art machine learning models for estimating forest AGB.

The improvements in RS technologies such as Sentinel-2 Multispectral Instrument, Sentinel-1 synthetic aperture radar and operational land imageries, and the open access policies provided immensely large sources of satellite data that enable the estimation of forest parameters including AGB (Madundo et al., 2023). The growing availability of multispectral satellite data together with the sophistication of programming language offers a huge capacity to predict forest biomass over a wide area with a little training dataset. Evidence in the field of machine

learning literature e.g. (Hastie, Tibshirani, Friedman, & Friedman, 2009) pointed out a variety of effective methods that enable the prediction of outcomes such as forest biomass, vegetation density, population, etc. as a function of features in a similar context or environment.

Machine learning algorithms such as LASSO (Robert, 1994), regression trees (Breiman et al., 1984), random forests (Breiman, 2001), support vector machines (Vapnik, 1999), etc. involve establishing a model of features and outcomes, with a penalty parameter for model complexity (Athey & Imbens, 2019). The linear regression-based machine learning model has the advantage of ease of implementation and interpretation, hence it is widely used in modeling AGB and other forest structure attributes using remotely sensed data (Omar et al., 2017). However, studies such as (Madundo et al., 2023; Tapamo et al., 2014) compared linear and nonlinear techniques and pointed out that linear regression models have limited ability to capture the complexity and non-linear interactions between AGB and RS variables. Lu et al. (2016) further highlighted that in forest biomass with high density and multi-layer canopy forest, RS data saturation occurs at AGB values between 100 and 150 Mg per ha. Their findings show a plateauing relationship between the RS information and forest biomass. As a result, the adoption of nonparametric statistical techniques such as generalized additive models and algorithms such as extreme gradient boosting (XGB) and light gradient boosting (LightGB) models, which can capture the non-linear and dynamic relationships between features and provide more accurate AGB estimates (Halperin et al., 2016; Soriano-Luna et al., 2018; Urbazaev et al., 2018) is increasing. Despite their wider use on temperate forests, studies on the application of these linear or non-linear machine learning tools to tropical forests, especially those dry afro-montane tropical forests in Africa, is limited (Madundo et al., 2023). As tropical trees also exhibited a decline in growth and photosynthesis at high temperatures, exploring the suitability of different modeling approaches in this unique environment is appropriate (Mau et al., 2018). Identifying relevant features from RS images and developing appropriate machine learning estimation techniques for specific studies remains far from adequate despite the progress in RS-based estimation over the past three decades. Parametric machine learning algorithms are based on the assumption of a constant/linear relationship between forest biomass and the associated features in the same environment, while non-parametric measures use small training data to enable biomass prediction with better accuracy (Wu et al., 2019; Zhao et al., 2019). The XGB regressor and LightGB regressor are improved algorithms based on gradient-boosting decision trees. Cross-validation in such cases is frequently used to determine the optimal complexity level for maximizing predictive power while avoiding 'overfitting'.

This study aims to estimate the aboveground biomass of dry afro-montane forests in developing countries using multispectral RS of multiple spectral bands of electromagnetic radiation captured by RS instruments with machine learning models trained based on inventory data from the Desa'a forest in Ethiopia. In this study, we focus on optimal hyperparameter tuning of different machine learning models to estimate aboveground biomass over a large area in dry afro-montane tropical forests using RS imageries.

2. Materials and Methods

2.1. Study Area and Data

This study is conducted based on RS images and ground-level inventory data collected from the Desa'a dry Afro-montane Forest in Northern Ethiopia, situated between in Tigray and Afar regional states. The forest is one of the largest and last remaining Afro-montane forests in Ethiopia (Bard et al., 2000; Haile, Birhane, Rannestad, & Adaramola, 2021). The forest covers an area of 120,026 (Hishe et al., 2015) ha and supports the livelihood of more than 250,000 people where over 26,000 people live below the poverty line and rely on the forest for water, energy, and to feed their cattle (WeForest, 2018). A large part of the forest is located in Tigray and is serving as a buffer and barrier against the expansion of the Denakil desert in the Afar region and highlands of the Tigray region (Gebreegziabher, 1999; Haile, Birhane, Mekonen Rannestad, & S. Adaramola, 2021). The forest is selected as a priority area for conservation by the government of Ethiopia since it is registered as one of the global biodiversity hotspot sites (Aynekulu, 2011).

Desa'a forest is also considered a candidate for REDD+ implementation because of its large size and the rapid increase in abundance, density, and aboveground biomass and thus carbon of the trees and expansive shrubs in the forest (Becknell et al., 2012). This necessitates the estimation of the aboveground forest biomass and carbon over the entire forest area in the region.

A. Inventory Data

The ground field inventory data on 253 field plots conducted from May 2016 to November 2017 by 'We Forest' are used for the calculation of AGB in this study (WeForest, 2018). The plots have an average area of approximately 0.04 ha (20 m x 20 m). AGB is computed based on the diameter at breast height (dbh) of individual trees and the diameter at stump height (DSH) of individual shrubs using the allometric model by Tetemke et al. (2019) developed with data from the study forest. There are different land use types within the forest boundary. The

inventory data used for the computation in this study, therefore, includes all kinds of land use in the forest boundary. This ensures the inclusion of various information necessary to train the model and estimate aboveground biomass of the entire forest. Finally, the computed AGB is converted to mg/ha. The mean AGB of plots in the area is 15.5 mg/ha with a range of 0.03 to 109.97 mg/ha and a standard deviation of 18.96 mg/ha.

B. Preprocessing and Remote Sensing Data Extraction

The remote sensing information used in this study come from Landsat 8 satellite images acquired in 2016 and resampled to 20 m pixel size so that it matches the field plot size. The study area is cloud-free region. After the image was processed using the Google Earth Engine (GEE) and the extraction of band reflectance values based on the forest plot locations is conducted using R codes. Finally, Python is used for generating vegetation indices from multispectral bands of atmospherically corrected RS information, feature engineering, feature selection, data analysis, and execution of the machine learning algorithms.

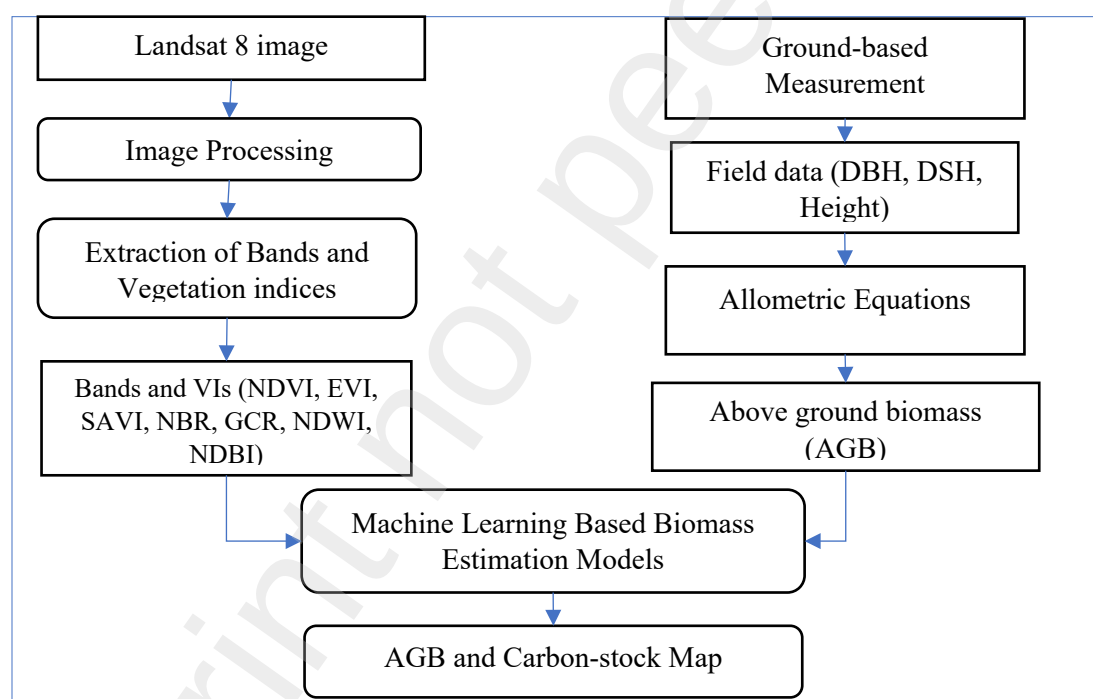


Fig 1: Methodological flow chart showing the data sources, analytical procedures¹, to the final output.

¹ For calculating AGB, we used two different equations developed by Tetemke et al., (2019) specifically for dry Afromontane forests of northern Ethiopia using data from Desa'a forest: $AGB = 0.298 \times DBH^{2.034}$ and for shrubs, $AGB = 0.17 \times DSH^{2.128}$ (Tetemke et al., 2019). The equation for trees and uses DBH in calculating AGB and the equation for shrubs uses DSH in the equation.

The vegetation indices (VI) used in the training and estimation of the machine learning models used this study includes Normalized Difference Vegetation Index (NDVI), Ratio Vegetation Index (RVI), Normalized Difference Built-up Index (NDBI), Normalized Difference Water Index (NDWI), and Soil-Adjusted Vegetation Index (SAVI), Structure Insensitive Pigment Index (SIPI), Normalized Burn Ratio (NBR), Green chlorophyll vegetation index (GCI), Moisture Stress Index (MSI) and Enhanced vegetation index (EVI). The list of indices and equations used for band arithmetic are presented in Table 1.

Table 1: List of vegetation indexes and equations used in this study.

Vegetation indices	Formula
Normalized difference vegetation index (NDVI)	$(\text{NIR}-\text{R})/(\text{NIR}+\text{R})$
Normalize Difference Built-up Index (NDBI)	$(\text{SWIR1}-\text{NIR})/(\text{SWIR1}+\text{NIR})$
Normalized Difference Water Index (NDWI)	$(\text{G}-\text{NIR})/(\text{G}+\text{NIR})$
Ratio vegetation index (RVI)	NIR/R
Soil-adjusted vegetation index (SAVI)	$(\text{NIR}-\text{R})/((\text{NIR}+\text{R}+\text{L}))*(1+\text{L}); \text{L}=0.5$
Moisture Stress Index (MSI)	$(\text{NIR}/\text{SWIR1})$
Normalized Burn Ratio (NBR)	$(\text{NIR}-\text{SWIR2})/((\text{NIR}+\text{SWIR2}))$
Green chlorophyll vegetation index (GCI)	$\text{NIR}/(\text{G}-1)$
Structure Insensitive Pigment Index (SIPI)	$(\text{NIR}+\text{B})/(\text{NIR}+\text{R})$
Enhanced vegetation index (EVI)	$2.5*(\text{NIR}-\text{R})/(\text{NIR}+6\text{R}-7.5\text{B}+1)$

2.2. Methods of Splitting the Dataset

The dataset for AGB was split using a K-Fold-Cross-Validation (K-Fold CV) for optimal model implementation. This strategy divides the statistical sample into separate subsets and then performs an analysis on part of them called the training set, while the so-called test sets are used to evaluate and check the reliability of its results (Ahmad et al., 2021; Weleszczuk et al., 2022). In this study, the dataset was divided into five sections. One section of data is used to test the model and the rest is used to train the model in five repetitions. This procedure continued until each of the five parts was used as the validating/test set, ensuring that each data point in the study was validated against each other.

2.3. Description and Evaluation of Machine Learning Models

The machine-learning algorithms used for the estimation of forest AGB are implemented with Scikit-learn in Python. Scikit-learn library is an open-source machine-learning package that includes algorithms to conduct classification, regression, and clustering (Pedregosa et al., 2011). The analysis started with simple linear regression (LR) to fit the dataset to a linear model that minimizes the sum of squares of residuals between the observed values of the data and the predicted values from the linear approximations, and then proceed to a more complex ensemble

techniques such as random forest, light gradient boosting, extreme gradient boosting regressors that capture the complexity of the non-linear interactions between features and forest AGB, i.e., the target variable.

Decision tree (DT) algorithms are used to build a model that learns simple decision rules from data attributes to predict the value of a target variable, and the prediction outcomes using DT algorithms are typically represented as decision tree grafts. Random forest (RF), similarly, is an ensemble technique based on bagging (Bootstrap Aggregating) and builds multiple DTs by bootstrapping the dataset and selecting random subsets of features for each tree. It then combines the predictions of these trees through voting or averaging. It is a meta-estimator that fits numerous DT algorithms (classifiers or regressors) on different sub-datasets of the examined dataset and forecasts accuracy using averaging methods. By controlling the model's over-fitting, RF selects the best regressors. RF is a robust and reliable algorithm that performs well on a wide range of tasks. It also typically has a lower variance and is less prone to overfitting. Over-fitting occurs when the developed model predicts the target variable on the training dataset too correctly and fails to accurately capture the real relationships between the target variable and features in the test dataset (Dunstan et al., 2022). Each tree is trained independently, so they tend to diversify and compensate for each other's errors. Regression trees differ from most other methods in the machine learning literature in that they produce a division of the population based on covariates, with all units in a partition receiving the same prediction (Hayes et al., 2015).

In Support Vector Regression (SVR), the main idea is to find a hyperplane that can be either linear or non-linear, depending on the kernel function, that best fits the data while also allowing for some margin of error. SVR can be configured as either a linear or a non-linear model depending on the choice of kernel function as it is not inherently a linear model. The proper setting of SVR meta parameters, the loss function, and the error penalty factor are key elements for the quality of SVR estimation. This study also used gradient boosting (GB) techniques that returns the fitted model as a set of weak models, which are typically DTs (Hastie, Tibshirani, Friedman, et al., 2009). XGB and LightGB regressors are extensions of GB applied in this study. XGB regressor is a model with numerous features for efficiently and flexibly processing missing data and assembling weak prediction models to produce an accurate model (Hou et al., 2020). For that purpose, XGB has many hyperparameters to fine-tune, which allows for more precise control over the model's behavior but may require more effort in hyperparameter tuning. XGB often achieves superior performance compared to Random Forest, especially in

structured/tabular data competitions on platforms like Kaggle despite being prone to overfitting. This technique is also known for its speed and efficiency in handling large datasets.

LightGB is also a boosting DT tool with a fast training speed, less memory usage, and better model performance which was established by the Microsoft DMTK team with a framework that implements the idea of gradient boosting decision tree algorithm (Ke et al., 2017). Ke et al. (2017) state that LightGB carries out Gradient-based One-Side Sampling and Exclusive Feature Bundling optimizations on the traditional GBDT algorithm. Gradient-based One-Side subsampling technique is used to create training sets for the base tree in the integration and selects data with larger gradients from sample to increase their contribution to the computed information gain while, Exclusive Feature Bundling merges specific data features to reduce the dimensionality of data (Su et al., 2021). In general, the hyperparameters used in the model greatly influence the prediction performance (Sun et al., 2020). Therefore, it is crucial to figure out the number of hyperparameters and how widely they may vary before using the LightGB regressor. Some of the important factors are the number of Leaves, Learning Rate, and Maximum Depth. The study also used a neural network (NN) model to predict the target variable based on the neural processing of data. The structure and principles of these specialized machine-learning models are inspired by the operation of a real (human) nervous system.

Google Collab and Jupiter notebook platform based on the Python 3.9.7 programming language are used to experiment with all the models in this study.

This study applies a K-fold cross-validation approach and splits the data into 5 datasets in order to evaluate the performance of several models used in the study: LR, SVR, RF, LightGB, and XGB regression models and implement three error metrics, namely, coefficient of determination (R^2) for the linear models, mean absolute error (MAE) and root mean square error (RMSE). The root mean square error (RMSE) is another method for assessing the performance of all models. In general, a higher R^2 value and lower MAE, RMSE and CV-RMSE values indicate a better estimation performance of models. R^2 is used to make a performance comparison among the linear machine learning models while RMSE is model agnostic metrics and used for the comparison of all linear and non-linear regression-based machine learning models.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \dots \dots \dots (1)$$

$$R2 = 1 - \frac{\sum_i^n (y_i - \hat{y}_i)^2}{\sum_i^n (y_i - \bar{y}_i)^2} \dots\dots\dots (2)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \dots\dots\dots (3)$$

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \text{ and } \epsilon_i = \sum_i^n (y_i - \hat{y}_i)^2$$

Where $\hat{y}_i$, $\bar{y}_i$ and y_i are the predicted forest AGB, mean forest AGB, and observed forest AGB values, respectively, whereas n is for the number of plots (data points). In essence, $\hat{y}_i$ is the predicted AGB from the models, y_i is the actual AGB value from the sample field inventory data, and n is the total number of sample plots included in this study.

2.4. Tuning and Optimization of Model Hyperparameters

This study also conducted the hyperparameter tuning on different machine learning methods used to estimate AGB and compared the predictive performance of the methods. In the estimation process, the sample dataset is first split into training and test datasets. A training dataset is a subsample used to train the regression model and estimate its parameters, while the test dataset is a subsample used to predict targets based on the trained models and evaluate the performances of models.

In this study, different alternative parameter values are specified for each parameter and a grid search method is implemented to try different combinations of hyperparameters and find the best hyperparameters for LightGB and XGB regressors. The range of hyperparameter values to search over is defined by the 'param_grid' dictionary which contains a list of values for: 'learning_rate', 'n_estimators', 'max_bin', 'bagging_fraction', 'num_leaves', 'bagging_freq', 'feature_fraction', 'min_sum_hessian_in_leaf' that are considered important while others not included in the parameter grid are taken at their default values. The grid search loops over each combination of hyperparameters and performs cross-validation to evaluate the model's performance.

For each hyperparameter combination, the code initializes an empty list to store the results of each cross-validation fold. It then loops over the cross-validation folds to fit the model to the training data and evaluates it on the validation data. The mean evaluation metrics across all folds are then computed and appended to the results list, along with the hyperparameter values. Grid search cross-validation and randomized search cross-validation techniques are used for hyperparameter tuning in this study.

Table 2: Parameter grid, hyperparameters, optimal parameters of each machine learning model.
NB: we set hyperparameters not listed in this table at their default values.

Model	Hyperparameter	Optimal hyperparameter
DT	'max_depth': [3, 5, 7]	3
	'min_samples_split': [2, 5, 10]	10
	'min_samples_leaf': (1, 2, 4]	2
RF	'n_estimators': randint [50, 500, 1000]	1000
	'max_depth': randint[2, 20]	2
	'min_samples_split': randint[2, 10, 20]	2
	'min_samples_leaf': randint[1, 5, 10]	1
	'max_features': ['auto', 'sqrt']	'sqrt'
GB	'n_estimators': [100, 200, 500]	200
	'learning_rate': [0.01, 0.1, 0.2]	0.01
	'max_depth': [3, 4, 5]	3
	'min_samples_split': [2, 3, 4]	4
	'min_samples_leaf': [1, 2, 3]	1
	'max_features': ['auto', 'sqrt', 'log2']	'auto'
LightGB	num_leaves': [6, 8, 10]	6
	learning_rate': [0.01, 0.05, 0.1]	0.01
	'n_estimators': [500, 1000, 2000]	500
	'max_bin': [100, 200, 300]	200
	'bagging_fraction': [0.6, 0.8, 1.0]	0.6
	'bagging_freq': [4, 6, 8]	8
	'feature_fraction': [0.2, 0.4, 0.6]	0.4
	'min_sum_hessian_in_leaf': [5, 10, 15]	5
XGB	'max_depth': [3, 5, 7]	5
	'learning_rate': [0.01, 0.05, 0.1]	0.05
	'n_estimators': [500, 1000, 2000]	500
	'subsample': [0.6, 0.8, 1.0]	0.6
	'colsample_bytree': [0.2, 0.4, 0.6]	0.2
	'gamma': [0, 1, 5]	1
	'min_child_weight': [1, 3, 5]	5
SVR	'regressor__C': [0.01, 0.1, 1, 10]	1
NN	learning_rate_init=0.06	0.01
	'activation': ['identity', 'logistic', 'tanh', 'relu']	tanh
	'solver': ['lbfgs', 'sgd', 'adam']	adam
	'alpha': [0.0001, 0.001, 0.01, 0.1, 1]	1

Grid search cross-validation method is used to improve model performances by tuning its hyperparameters using systematic ways to try different combinations of hyperparameters in the modeling and evaluate the model's performance using a cross-validation approach. In grid search CV, a grid of hyperparameter values is first specified to train the model and evaluate the model on each combination of hyperparameters in the grid. Therefore, the evaluation is typically done using k-fold CV, in which the data is split into k equally sized subsets, and the model is trained and tested k times, each time using a different subset for testing and the remaining subsets for training. The performance of the model on each combination of hyperparameters is then evaluated by computing a metric such as root mean square error or mean squared error, and the combination of hyperparameters that results in the best performance is selected as the optimal hyperparameters. Grid search CV is thus, a powerful method to find the best hyperparameters of a model, but it can be computationally expensive, especially when the grid of hyperparameters is large or when the dataset is large. This study used the technique to improve the performance of the machine learning models.

Randomized search CV technique is similar to grid search cross-validation, but instead of trying all combinations of hyperparameters, it randomly samples a subset of hyperparameters from the search space. Randomized search CV is computationally more efficient than grid search in a situation where the search space is too large. In some cases, when the search space is small or when certain hyperparameters are known to be more important than others, domain knowledge or prior experience with similar models can be used to guide the selection of hyperparameters.

Bayesian optimization is an iterative method that uses Bayesian inference to construct a probabilistic model of the performance as a function of the hyperparameters and use this model to select the next hyperparameters to try, intending to find the optimal hyperparameters with as few evaluations as possible. In gradient-based optimization, the hyperparameters are treated as variables to be optimized using gradient descent or a similar optimization algorithm, and this can be effective when the performance of the model is differentiable concerning the hyperparameters.

Overall, the choices of hyperparameter tuning are determined by the type and specificities of the problems and the available resources, and different methods may be more or less effective in different contexts. In contrast, model validation is overlooked and receives only little emphasis in most discussions of linear regressions in econometric estimation. For example, Athey and Imbens (2019) highlighted that the structure of the regression models, parametric or nonparametric, and the set of regressors are presumed to be externally defined for instance, by

economic theory. Given this specification, the task for the researchers is to estimate the unknown parameters of the model. There is a significant emphasis on conducting efficient estimation, and it is often operationalized through definitions of large sample efficiency. The discussion of model selection also typically involves testing null hypotheses related to the validity of a specific model, implying the existence of a true model that should be chosen for subsequent tasks. Two key aspects contribute to this capability. Firstly, the primary objective is predictive power instead of the estimation of a particular causal or structural parameter, and secondly, the approach involves out-of-sample comparisons instead of in-sample goodness-of-fit measures.

3. Results and Discussion

The sample plots in this study include plots from dense forests, open forests, shrublands, exclosures, pasturelands, and settlements in the estimation of aboveground biomass. The highest aboveground biomass is registered from the plots in dense forests and open forests, respectively while the highest variation manifested in the coefficient of variation is observed in exclosures and settlement areas.

3.1. Assessments of Model Performances

Using the R² and RMSE values as metrics (see Table 2), we evaluated the performances of the models at each stage with the test data from the corresponding stages and compared these models to determine and select the one with the highest prediction performance. For each stage, the R² and RMSE values between estimated forest AGB from the models and observed forest AGB values are also presented graphically using scatter plots.

Table 3: Overall performance evaluation results of linear and ensemble models with moderate (randomized search cross-validation) hyperparameters tuning

Model	MAE	MSE	RMSE	R2 Score	CV-RMSE
Neural Network Regressor	0.3308	0.1871	0.4327	-	-
Random Forest Regressor	0.3349	0.1883	0.4340	0.1881	0.4371
Linear Regressor	0.3347	0.1884	0.4340	0.1880	0.4363
Ridge Regressor	0.3298	0.1892	0.4350	0.1845	0.4258
XGB Regressor	0.3507	0.1958	0.4425	0.1558	0.4323
SV Regressor	0.3165	0.1966	0.4434	0.1524	0.4180
LightGB Regressor	0.3517	0.1973	0.4442	0.1494	0.4230
ElasticNet Regressor	0.3490	0.1990	0.4461	0.1422	0.4137
LASSO Regressor	0.3655	0.2074	0.4554	0.1062	0.4188
Polynomial Regression (degree=2)	18.3001	2305.823	48.0190	-9938.897	0.4363

Table 3 presents the MAE, MSE, RMSE, and CV-RMSE of all machine learning models used in this study. The RF, LightGB, and XGB in Table 2 are with minimal hyperparameter tuning. The evaluation results indicate that the models reveal somewhat similar performance, and there are no significant differences between the models. All the models exhibit RMSE higher than 0.43 and R2 lower than 0.19.

On the other hand, optimization of the hyperparameters for the higher-level ensemble models such as RF, LightGB, and XGB regressors with the gross search CV as indicated in Table 4, the models demonstrate an overall improvement in the predictive performance in comparison to SVR and the linear regression-based machine learning models (LM, LASSO, Ridge, Elastic Net) and all models with randomized search CV. The RMSE of the RF, LightGB, and XGB regressors decreased to 0.235, 0.243, and 0.250, and the corresponding R2 values increased to 0.945, 0.941, and 0.938 respectively. These results are higher compared with SVR, linear regression-based machine learning models such as LASSO, ridge, and Elastic Net models (see Table 3).

Table 4: RMSE and R-squared computed from the random forest, LightGB, XGB, and GB regressors with (gross search CV) optimal hyperparameters tuning.

Machine Learning Models	RMSE	R2 Scores
RF regressor	0.235	0.945
LightGB regressor	0.243	0.941
XGB regressor	0.250	0.938
Gradient Boosting regressor	0.331	0.891

Regression trees and their extension, random forests are widely used and effective application methods for flexible estimation of regression function in settings where out-of-sample prediction is important. RF regressor, without requiring subtle tuning, exhibits high out-of-the-box performance. The algorithm in the random forest takes the training sample for example, X_{ik} , Y_i and split it into subsamples to estimate regression results within the subsamples and the average value of sub-sample estimates (Athey & Imbens, 2019). The model starts with the full training sample, and step by step continues to split based on some features or covariates and thresholds. The random forest algorithm is based on a single covariate x_{ik} at a time exceeding a threshold c and results in a reduction in model variance.

The results in Table 3 show that even with moderate hyperparameter tuning, RF regression gives better out-of-the-sample performance compared to gradient boosting, XGB, and LightGB regressors in all evaluation metrics. This indicates the relatively less susceptibility of RF to

overfitting. The MAE and RMSE results also reveal that neural network models exhibit better performances in predicting the outcomes. The results reveal that the performance of RF regressors, LightGB regressor, and XGB regressor is highly improved with grid search cross-validation technique in tuning hyperparameters compared to with the use of randomized search cross-validation which indicates moderate hyperparameter tuning. The use of gross search CV for hyperparameter tuning could be important especially when the data is limited.

3.2. Feature and permutation importance

In this study, we first included features that have a correlation coefficient of 0.25 or above with the target variable and conducted the feature importance scores and permutation importance scores in the estimated machine learning models. L1 regularization (e.g., lasso regression), and shapely values are also widely applied methods to examine and compare variable importance. Feature importance scores are common for tree-based models (e.g., Decision Trees, Random Forests, Gradient Boosting) and indicate how much each feature contributes to reducing the model's impurity, e.g., Gini impurity in decision trees. On the other hand, permutation importance measures the change in model performance (e.g., root mean square error or mean squared error) when you randomly shuffle the values of a single feature, and a larger drop in performance indicates that the feature is more important. L1 regularization is based on the size of the estimated coefficient and is relevant for linear models while a model-agnostic technique assigns a value to each feature, representing its contribution to the prediction. Since these techniques have different parameters and algorithms, the variable importance score results from these techniques are different. The results from feature importance scores and permutation importance scores are presented in Figure 2 below. The variable importance results for all models are also presented in Appendix II.

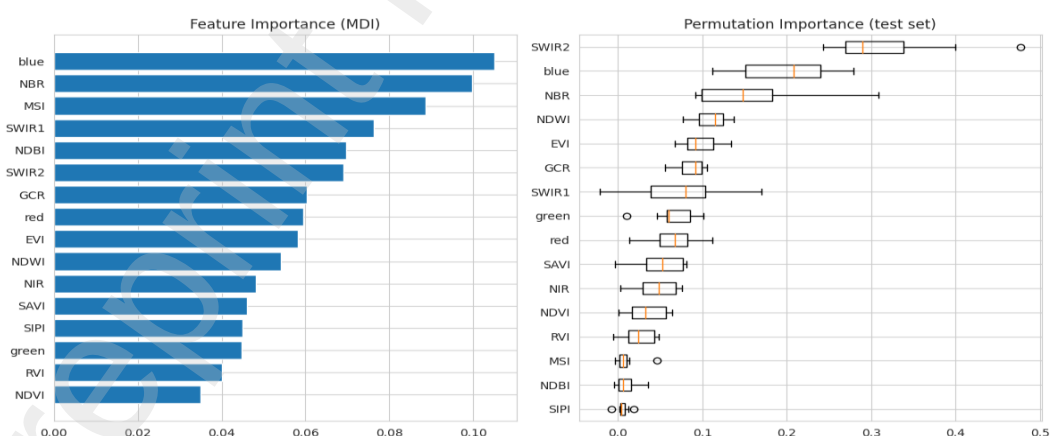


Figure 2: Feature importance score and Permutation importance scores

The results also show that mean differences between actual values (with a mean of 1.39) and predicted values (with a mean of 1.33) are not statistically significant for the selected models. In addition, we assessed the correlation between the actual and predicted values of the target variable to evaluate these models and found correlation coefficients greater than 0.86 with a $p > 6.11e^{-14}$ (see Table 3). These tests indicate the suitability of the models in predicting the forest AGB using the RS information.

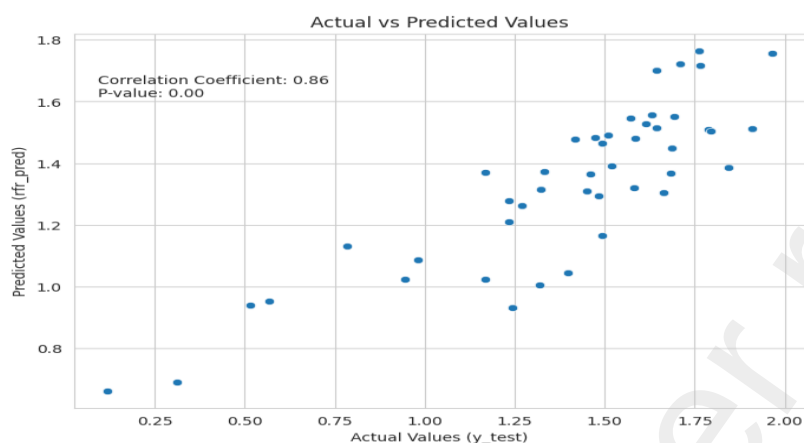


Figure 3: The correlation coefficient and p-values of of the predicted and actual target outcome.

3.3. Hyperparameter Tuning, Loss function, and Regularization

The results from LightGB and XGB regressors are used to predict and map the forest AGB in this study since they demonstrate better performances compared to the other linear and nonlinear models. LGBR and XGB models with little tuning of hyperparameters result in lower R^2 (0.26) and lower prediction performance compared to the other models. The inclusion of hyperparameters tuning however improved the performance and results in a reduction of RMSE from 1.8 tone/he to 0.250 tone/he and an increase of R^2 from 0.26 to 0.945, which shows a significant increase in the explanatory power of the models to explain the variations in the AGB values. The best hyperparameters for the final LightGB model from the grid parameters are `bagging_fraction`: 0.6, `bagging_freq`: 8, `feature_fraction`: 0.4, `learning_rate`: 0.01, `max_bin`: 200, `min_sum_hessian_in_leaf`: 5, `n_estimators`: 500, `num_leaves`: 6.

We use the binomial deviance loss function to test how regularization affects the predictive performance of models. The results in Figure 4 show that regularization via shrinkage² (learning

² "shrinkage" and "no shrinkage" refer to the choice of whether to apply a learning rate (shrinkage) less than 1 or not (learning rate equal to 1) in gradient boosting algorithms. Shrinkage is a regularization technique that helps control the contribution of individual trees, leading to better generalization, while no shrinkage allows for faster convergence but increases the risk of overfitting. The size of the dataset, and the desired trade-off between training speed and model generalization determine the

rate < 1) improves the performance considerably and combining stochastic gradient boosting (subsample < 1) with shrinkage produces more accurate models by reducing the variance via bagging. However, according to Hastie, Tibshirani, Friedman and Friedman (2009), subsampling without shrinkage often performs poorly. Figure 4 presents the mean squared error for different stages (iterations) in the gradient boosting model and training set deviance³.

Training and test deviances with number of iterations Test deviance and regularizations of different parameters

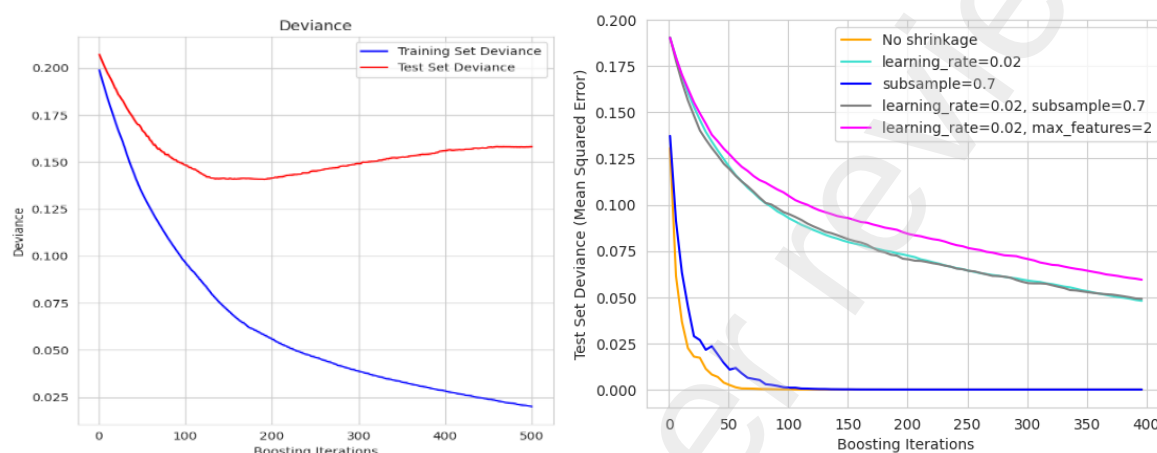


Figure 4: Illustration of the effect of different regularization strategies for Gradient Boosting DT algorithms can often be complex or prone to overfit. It is important to implement the right set of hyperparameter adjustments based on the availability of data and various statistical metrics that give us an overall understanding of how the data is. We conducted an optimization with grid search and random search techniques to come up with the optimal hyperparameters for RF, XGB, and LightGB models. We also tested the importance of hyperparameters set in the models we use for AGB estimation. The choice of hyperparameters is tuning them to improve the predictive performance of the models and prevent overfitting and underfitting problems. The least-square function is used to optimize the loss function. The training deviance and test deviance are indicators to decide the number of boosting indicators. It is important to note the test deviance is U-shaped indicating it has a minimum value and the training deviance can decline indefinitely. This implies that the model can perform better on the training data its mean squared error declines while it performs badly on the test data. Thus, it is important to optimize the loss function on test deviance to prevent underfitting and overfitting problems.

choice of these two approaches. It's common to tune the learning rate along with other hyperparameters to find the best balance for the task.

³ Training set deviance refers to as the "training deviance" or "training loss," is a metric that measures how well a gradient boosting model fits the training data as the number of boosting iterations (trees) increases. It is used to monitor the model's convergence during training.

The baseline requirement of resources to train a complex model: the size of data, and the time it takes to train the algorithm are crucial factors involved in the experimentation phase when working with machine learning models. In this study, the RF regressor is found to be faster and takes less time than LightGB and XGB regressors to finish the training of the model and performs better in comparison. Despite the close results from LightGB and XGB regressors, the later needs more estimation time (3 hours and 46 minutes) comparatively.

Therefore, with the increasing availability of RS imagery and cloud computing technology coupled with machine learning models for analyzing big data, models trained based on local data can offer an immense opportunity for a least cost, accurate and faster ways of estimating aboveground forest biomass and thus carbon, especially in developing countries where the need for forest monitoring is high and forest are fragmented and widely distributed over a large areas (Forkuor et al., 2020; Fremout et al., 2022). The relevance of such studies is even higher for poor developing countries like Ethiopia, in which high resource constraints and frequent conflicts limit fieldwork surveys. This study established different machine learning models including linear, lasso and ridge, support vector regressor, random forest regressor, extreme gradient boosting, and light gradient boosting regressors to better estimate the aboveground biomass for dry afro-montane forest in North Ethiopia and compared the predictive performance of the models. We use information on multispectral indices, vegetation indices, and ground-level data to train, evaluate and compare the predictive performance of the models with evaluation metrics (MSE, MAE, RMSE, and R^2).

The findings show that random forest regressor, light gradient boosting regressor, and extreme gradient boosting regressor estimate AGB with 0.235, 0.243, and 0.250 RMSE respectively, and demonstrate better predictive performances relative to other models. Support vector regression also provides stable AGB estimation. We also computed feature and permutation importance for the models and found that all features are not consistently important in different models since the algorithms and hyperparameters vary across models. The results also show that hyperparameters tuning in the RF, LightGB, and XGB regressors greatly increase the predictive performance of these models and reduce the overfitting problems compared to default hyperparameters. In contrast, linear models also enable forest AGB estimation with few features and hyperparameters since these models in general exhibit lower prediction errors compared to ensemble models with modest tuning/regularization. The main problem with linear models is they assume a linear relationship between the target variable and features and tend to drop most of the features in the computation and attribute the end evaluation performance to a

few variables, i.e., linear models attribute all the predictive importance to a few variables and make predictions accordingly (see Annex III). Eventually, this study demonstrates that ensemble models outperform linear models with optimal hyperparameter tuning.

The findings in this study also display that the ranges of evaluation metrics highly vary for RF, LightGB, and XGB machine learning models due to several factors, including the model's complexity, sensitivity to outliers, and underlying assumptions compared to linear models. However, the use of complex ensemble models like RF, LightGB, and XGB with appropriate tuning of the hyperparameters can capture non-linear relationships and complex interactions between features, and are more sensitive to outliers, as they may put more emphasis on minimizing the most significant errors.

The other important reason is that ensemble models have fewer assumptions about the data, making them more flexible but also more prone to overfitting and a broader range of outcomes, especially when data quality is variable. The wider range of evaluation metrics for ensemble models in this study reflects their ability to adapt to complex relationships in the data, but it also requires careful hyperparameter tuning and model selection to mitigate overfitting. CV-RMSE is the RMSE calculated through cross-validation, and it assesses the model's generalization performance by averaging RMSE across multiple validation folds.

4. Conclusion

This study is conducted in a tropical dry afro-montane forest in Northern Ethiopia to estimate the AGB based on Landsat 8 satellite imagery using the applications of different machine learning algorithms including, LR, LASSO, Ridge, Elastic Net, SVR, RF, LightGB and XGB. The study has revealed some important findings. Hyperparameter tuning or parameter optimization with grid search cross-validation greatly improves the performance of machine learning models. The results demonstrate hyperparameter tuning significantly improves the performance of LightGB and XGB: tuning increases the R^2 of the LightGB regressor to 0.941 and the XGB regressor to 0.938. The results from the RF regressor are more consistent and improved predictive performance even compared to LightGB and XGB regressors. The RMSE of the RF, LightGB, and XGB are 0.235, 0.243, and 0.250 respectively. Comparatively, these models reduced the problem of underestimating high AGB values and overestimating low AGB values to a certain extent and made the predicted AGB results closer to the ground test data results. The RF, LightGB and XGB models with tuning significantly improved the predictive performance compared to the LR models. The study also shows that despite the time it takes,

the model performance with grid search CV in hyperparameters tuning is greatly improved compared to tuning with randomized search CV algorithm. This study used Landsat 8 imagery and multispectral indices to estimate the AGB of dry afro-montane forests. The combination of the Sentinel image with the Landsat 8 image and the inclusion of auxiliary data such as climate, soil, and DEM may help to improve model performance in regional biomass estimation. The prediction models from RF, XGB, and LightGB algorithms are scalable and can be used to predict the AGB of dry afro-montane forests in other developing country regions. These algorithms are particularly relevant since the alternative manual ground-based forest investigation consumes time and can sometimes be destructive and constrained in developing countries due to its high costs and incidence of conflict in these regions.

Credit authorship contribution statement.

Abrha Megos Meressa: Conceptualization, methodology, validation, formal analysis, visualizations, writing original draft, review & editing. **Meley Mekonen Rannestad:** conceptualization, writing – review & editing, project administration, and funding acquisition.

Conflict of Interests

The authors declare that there is no conflict of interest regarding the publication of this paper.

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Supplementary data

Supplementary data and code to this paper will be available online at github.

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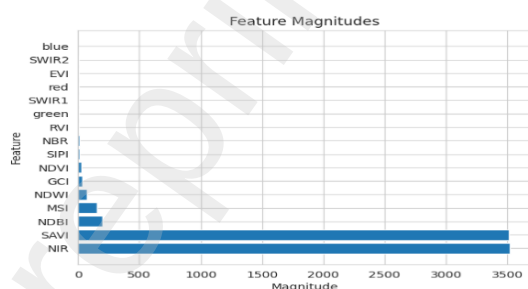
Annex

I. Correlation Matrix

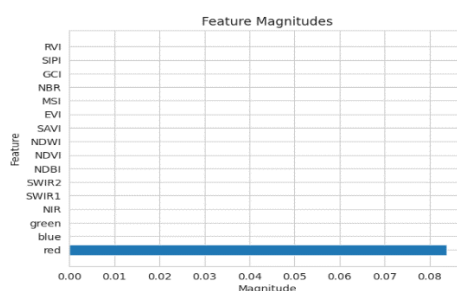
AGB_mean	blue	green	red	NIR	SWIR1	SWIR2	NDBI	NDVI	NDWI	SAVI	EVI	MSI	NBR	GCI	SIPI	RV1
1.00	-0.41	-0.42	-0.42	-0.37	-0.41	-0.36	0.42	-0.41	-0.26	0.41	0.35	0.41	0.41	0.40	0.42	
-0.41	1.00	0.98	0.97	0.71	0.84	0.88	0.64	-0.88	0.87	0.71	-0.85	-0.64	-0.73	-0.87	-0.86	-0.86
-0.42	0.98	1.00	0.99	0.78	0.88	0.91	0.63	-0.86	0.84	0.78	-0.87	-0.63	-0.72	-0.84	-0.81	-0.85
-0.42	0.97	0.99	1.00	0.77	0.90	0.94	0.68	-0.88	0.84	0.77	-0.90	-0.68	-0.76	-0.84	-0.81	-0.88
-0.26	0.71	0.78	0.77	1.00	0.86	0.81	0.29	-0.37	0.32	1.00	-0.47	-0.29	-0.35	-0.32	-0.28	-0.37
-0.37	0.84	0.88	0.90	0.86	1.00	0.98	0.74	-0.67	0.60	0.86	-0.74	-0.74	-0.76	-0.61	-0.57	-0.68
-0.41	0.88	0.91	0.94	0.81	0.98	1.00	0.78	-0.75	0.69	0.81	-0.81	-0.78	-0.84	-0.70	-0.66	-0.77
-0.36	0.64	0.63	0.68	0.29	0.74	0.78	1.00	-0.76	0.72	0.29	-0.77	-1.00	-0.96	-0.72	-0.71	-0.80
0.42	-0.88	-0.86	-0.88	-0.37	-0.67	-0.75	-0.76	1.00	-0.98	-0.37	0.97	0.76	0.84	0.98	0.98	0.99
-0.41	0.87	0.84	0.84	0.32	0.60	0.69	0.72	-0.98	1.00	0.32	-0.92	-0.72	-0.80	-1.00	-0.98	-0.97
-0.26	0.71	0.78	0.77	1.00	0.86	0.81	0.29	-0.37	0.32	1.00	-0.47	-0.29	-0.35	-0.32	-0.28	-0.37
0.41	-0.85	-0.87	-0.90	-0.47	-0.74	-0.81	-0.77	0.97	-0.92	-0.47	1.00	0.77	0.84	0.92	0.90	0.97
0.35	-0.64	-0.63	-0.68	-0.29	-0.74	-0.78	-1.00	0.76	-0.72	-0.29	0.77	1.00	0.96	0.72	0.71	0.80
0.41	-0.73	-0.72	-0.76	-0.35	-0.76	-0.84	-0.96	0.84	-0.80	-0.35	0.84	0.96	1.00	0.81	0.79	0.86
0.41	-0.87	-0.84	-0.84	-0.32	-0.61	-0.70	-0.72	0.98	-1.00	-0.32	0.92	0.72	0.81	1.00	0.98	0.97
0.40	-0.86	-0.81	-0.81	-0.28	-0.57	-0.66	-0.71	0.98	-0.98	-0.28	0.90	0.71	0.79	0.98	1.00	0.96
0.42	-0.86	-0.85	-0.88	-0.37	-0.68	-0.77	-0.80	0.99	-0.97	-0.37	0.97	0.80	0.86	0.97	0.96	1.00

II. Variable importance (Coefficient magnitudes) for Linear models

A. Linear regression

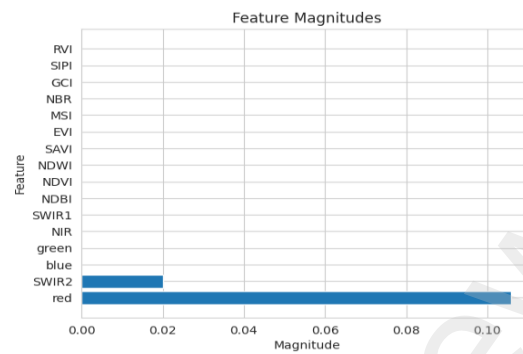
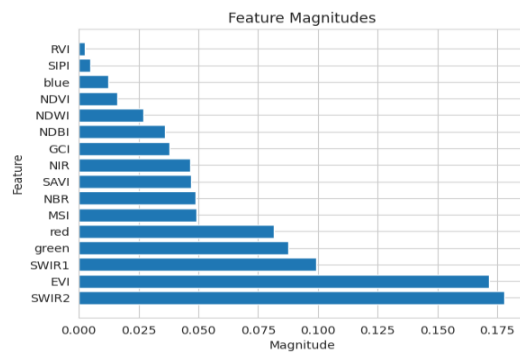


C. LASSO



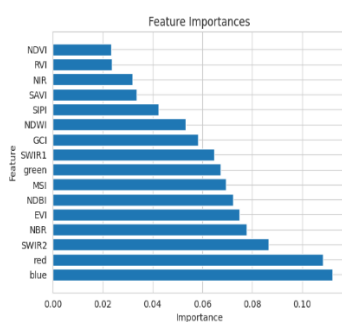
B. Ridge

D. Elastic Net

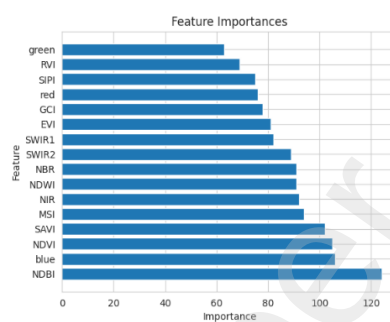


III. Feature importance for Ensemble models

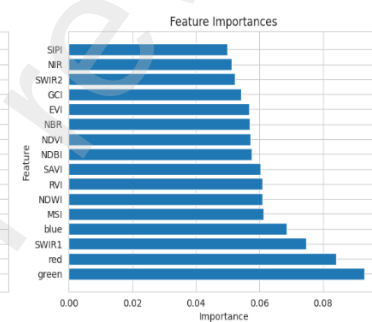
A. Random Forest regressor



B. LightGB regressor



C. XGB regressor



IV. The digestion and fitting line of predicted values and the actual values.

